Cyber Security Assessment Report

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DHC-826

Acknowledgment

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I am especially thankful to my Team Leader Faizan Khan, for their guidance, encouragement, and constructive feedback throughout the internship. Their mentorship has helped me gain deeper insights into cybersecurity practices and professional standards.

Disclaimer: "*This document has been prepared solely for learning purposes. It is intended to demonstrate knowledge and skills acquired during the internship. The content does not promote, encourage, or support any illegal, unethical, or unauthorized activity in the field of Cyber Security".*

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CYBERSECURITY REPORT (Week 1)

Title:

Security Assessment Report of Web Application

Objective:

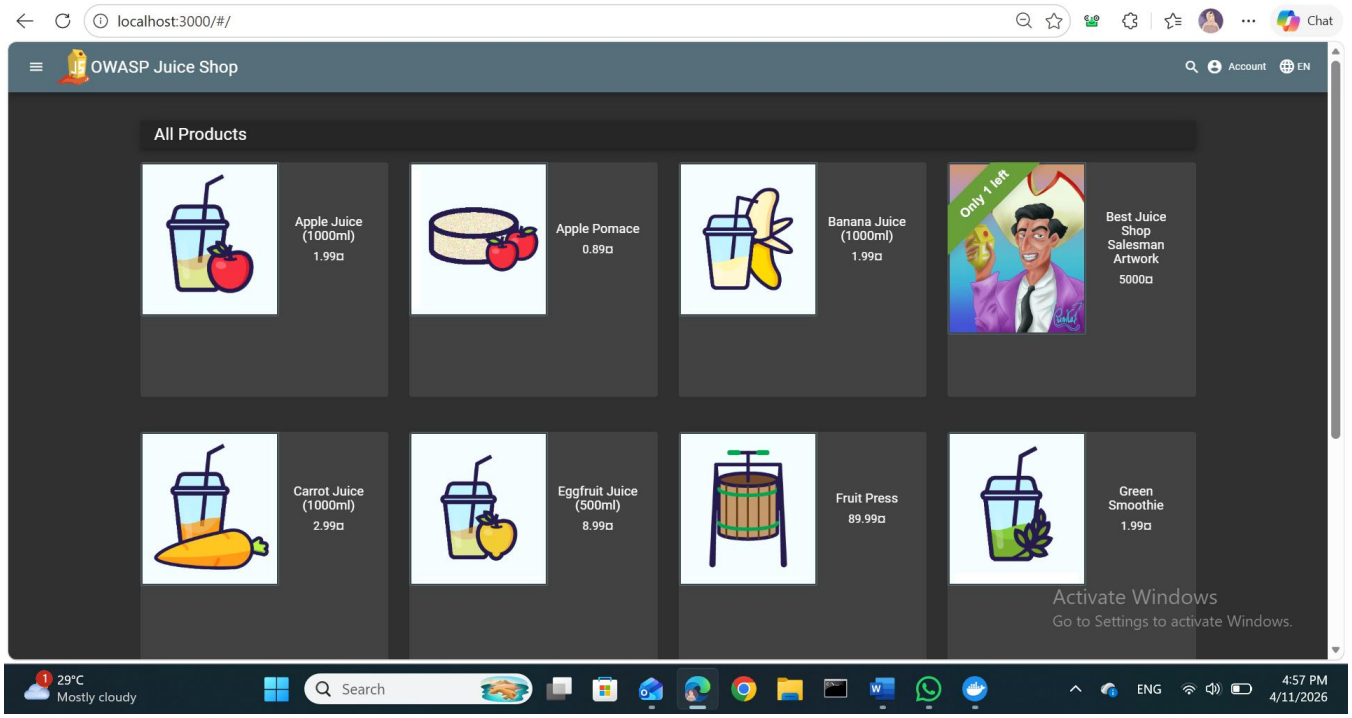
The objective of this task was to analyze a web-based application for common security vulnerabilities and identify areas for improvement using manual and automated testing techniques.

Application Setup:

- A vulnerable web application was set up using Docker.
- The application was accessed at:

  http://localhost:3000

- The application was explored including login, signup, and product review sections.



Vulnerability Assessment

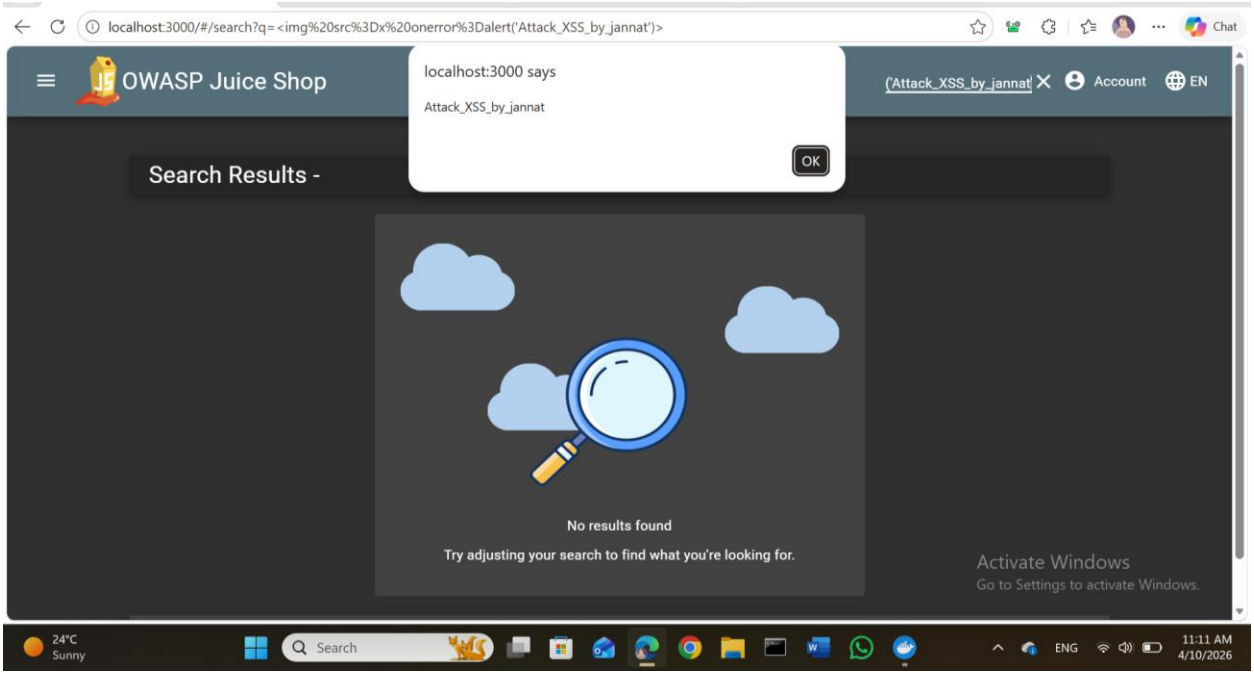
1. Cross-Site Scripting (XSS)

Reflected XSS:

- Payload used:

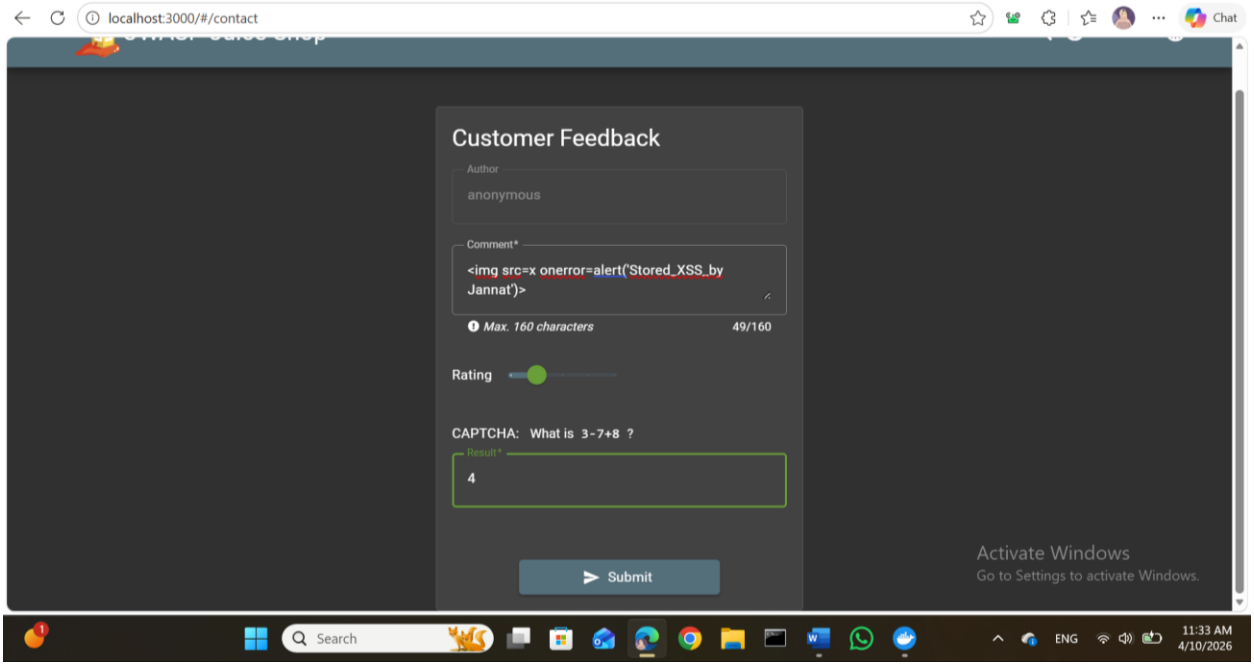
  <img src=x onerror=alert('Attack_XSS_by_jannat')>
- Result:
  A popup alert was triggered immediately after input.

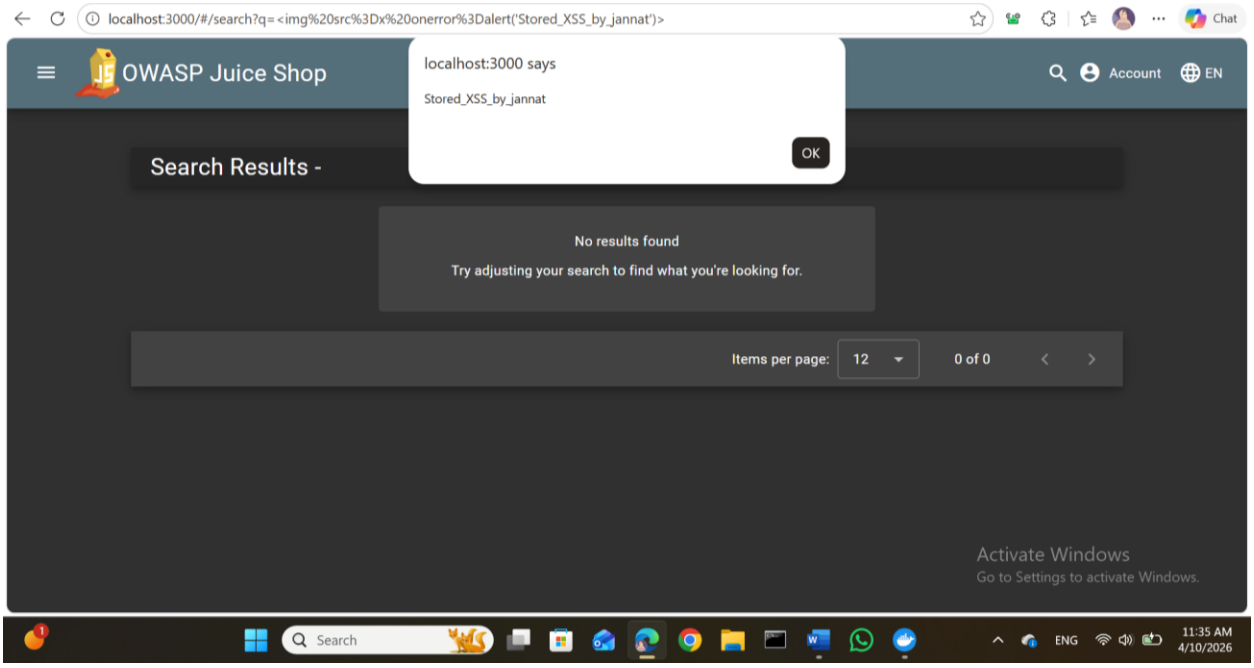
Vulnerability Confirmed



Stored XSS:

- Payload used:
  <img src=x onerror=alert('Stored_XSS_By_Jannat')>
- Location: Customer feedback section
- Result:
  The payload was stored and executed repeatedly on page reload.
- Vulnerability Confirmed



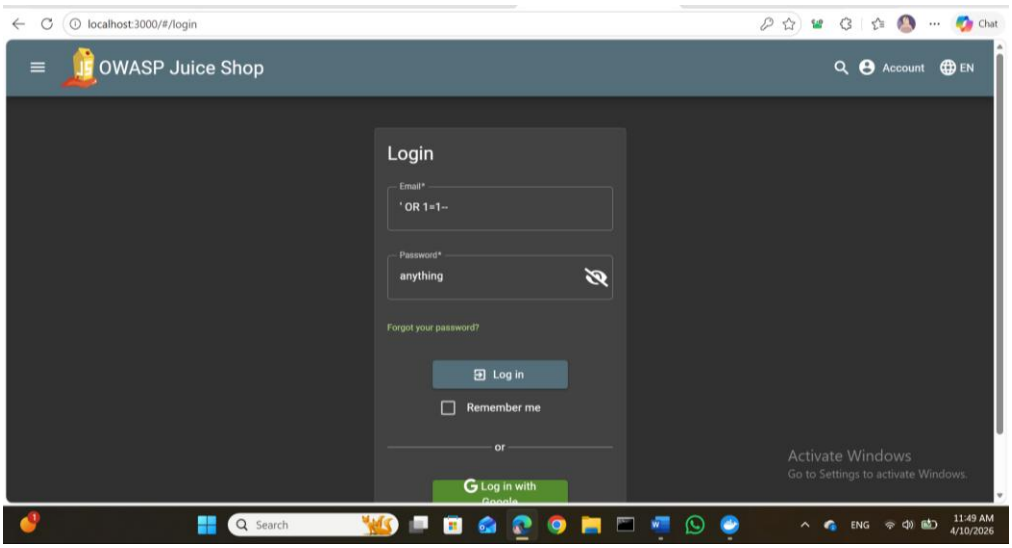


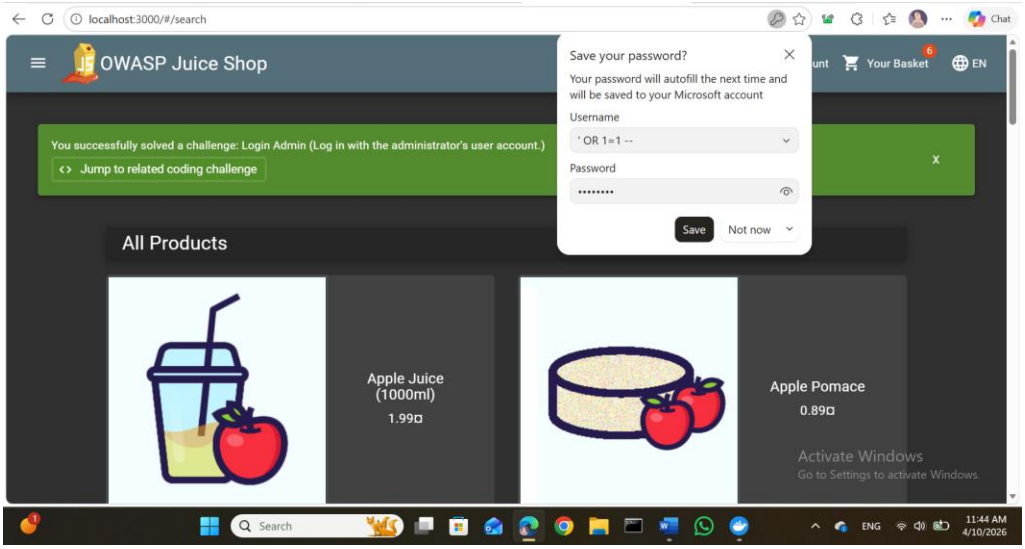
2. SQL Injection

Login Bypass:

- Payload used:
  ' OR 1=1 --
- Result:
  Login was successful without valid credentials.

Vulnerability Confirmed





Error-based Behavior:

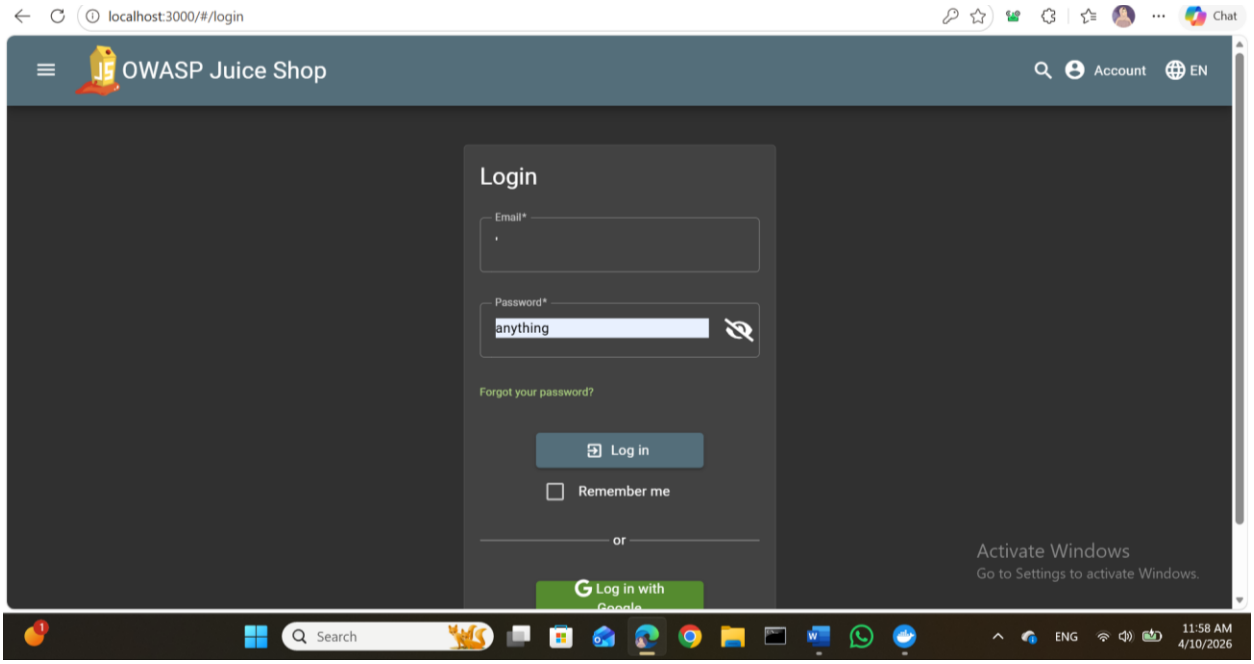
- Input used:

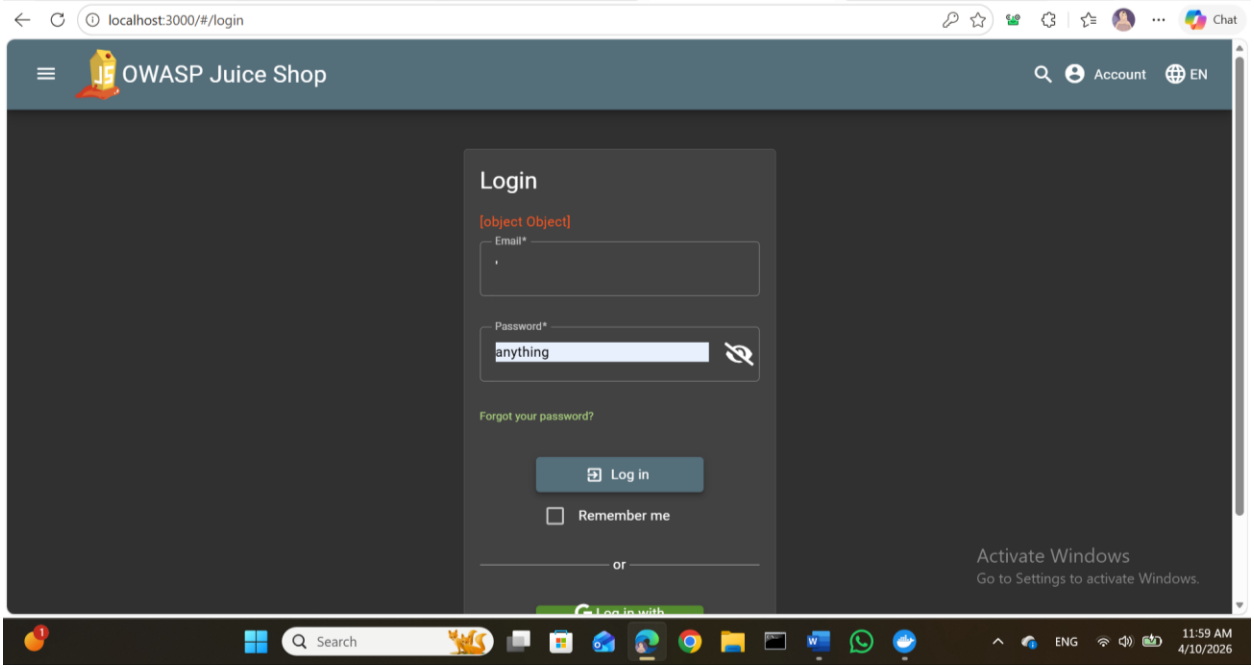
  '

- Result:

  The application returned "[object Object]", indicating improper error handling.

Potential Information Leakage

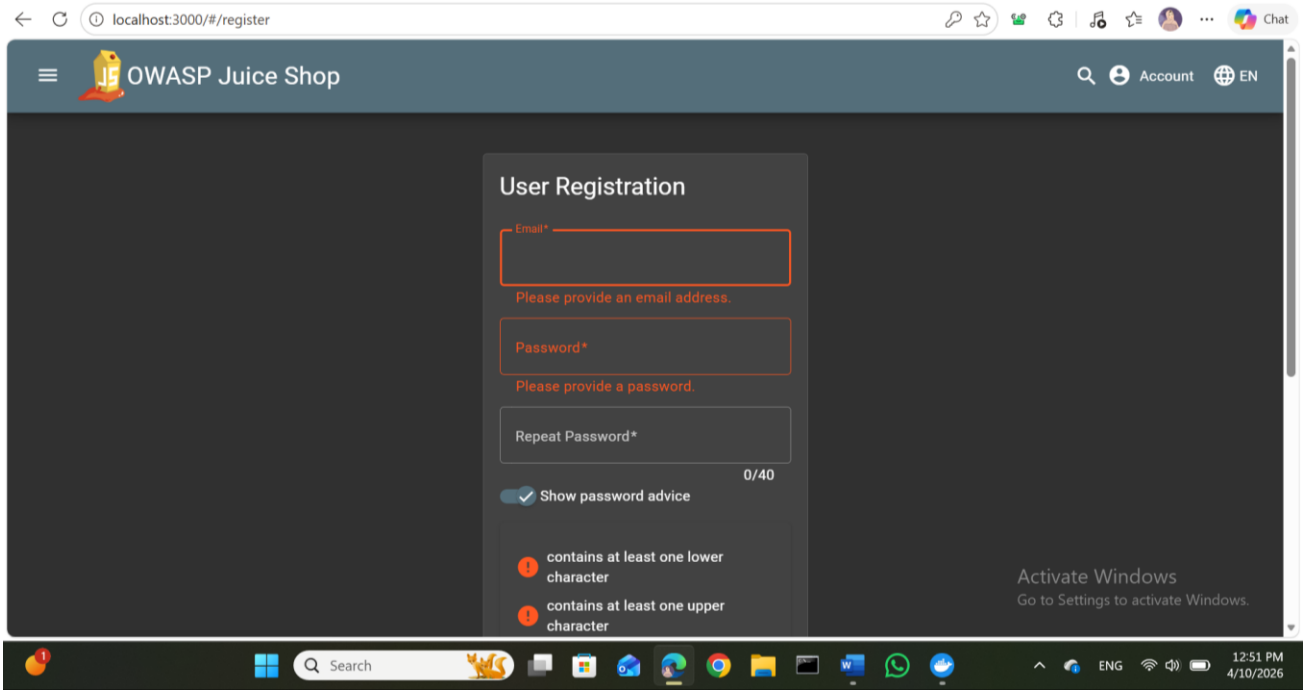


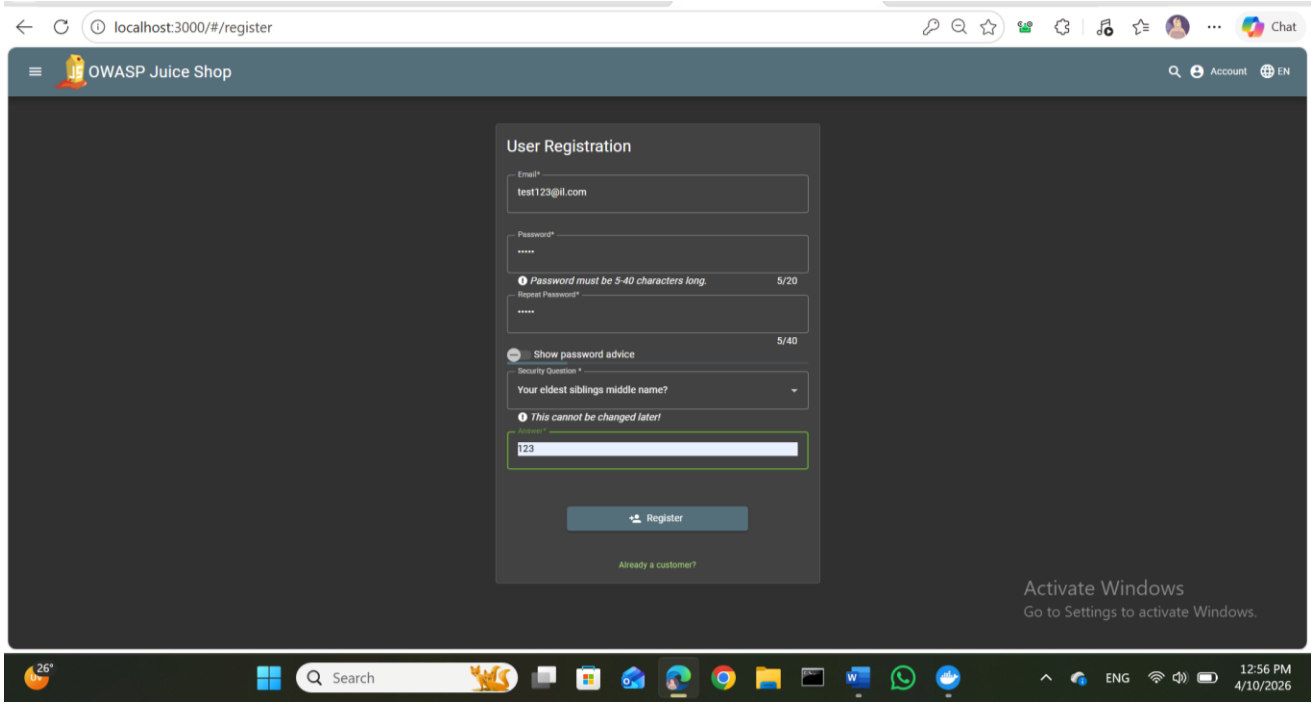


3. Weak Password Policy

- Weak password used: 12345
- Email used: test123@il.com
- Result:
  The application allowed account creation with a weak password.

No password strength enforcement

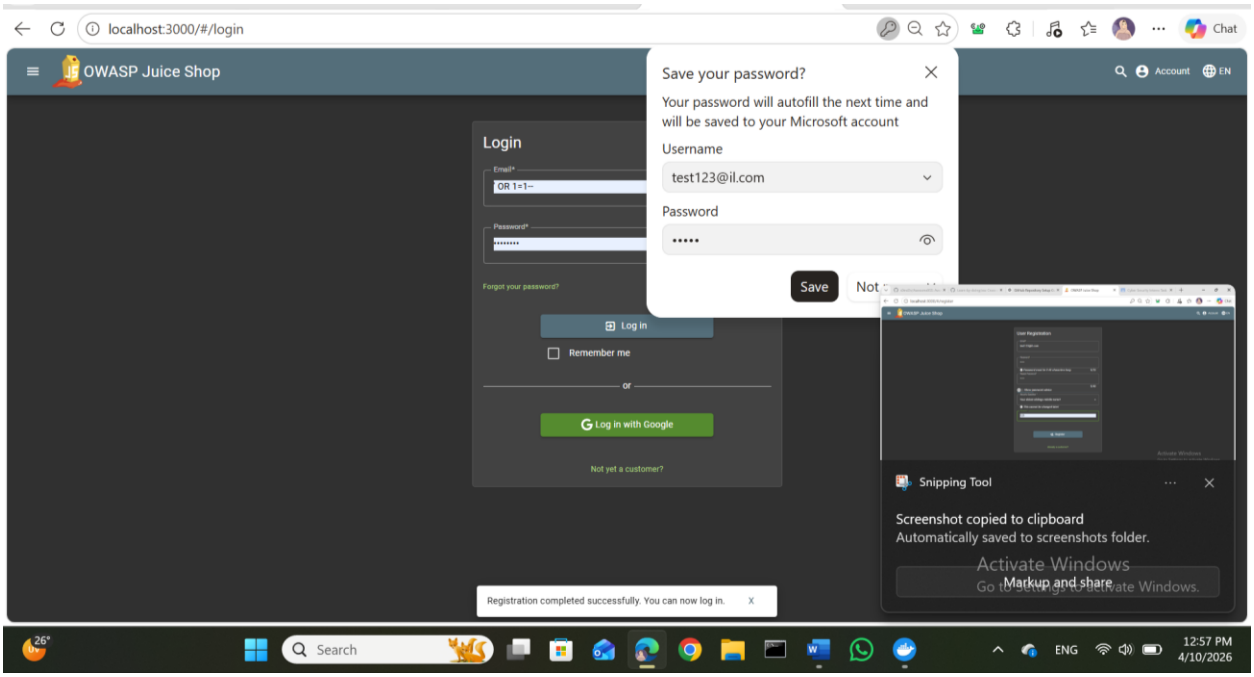




4. Input Validation Issue

- Email used: test123@il.com
- Result:
  The application accepted an invalid email format.

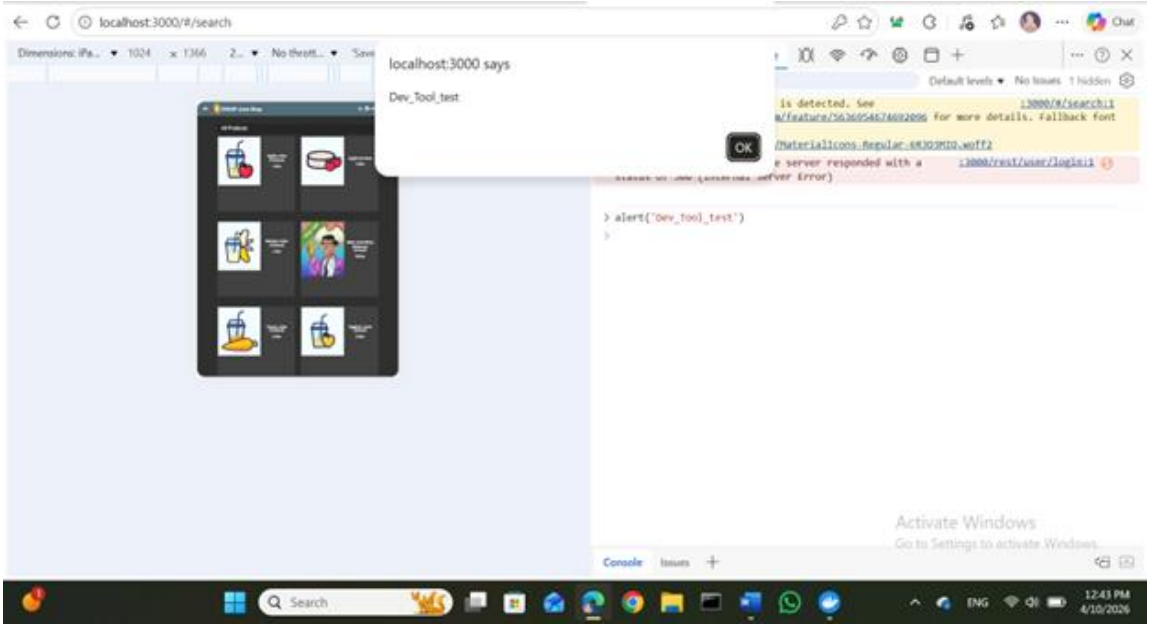
Improper input validation



5. Browser Developer Tools Testing

- JavaScript execution tested using:
  alert("Dev_tool_test")
- Result:
  The script executed successfully in the browser console.

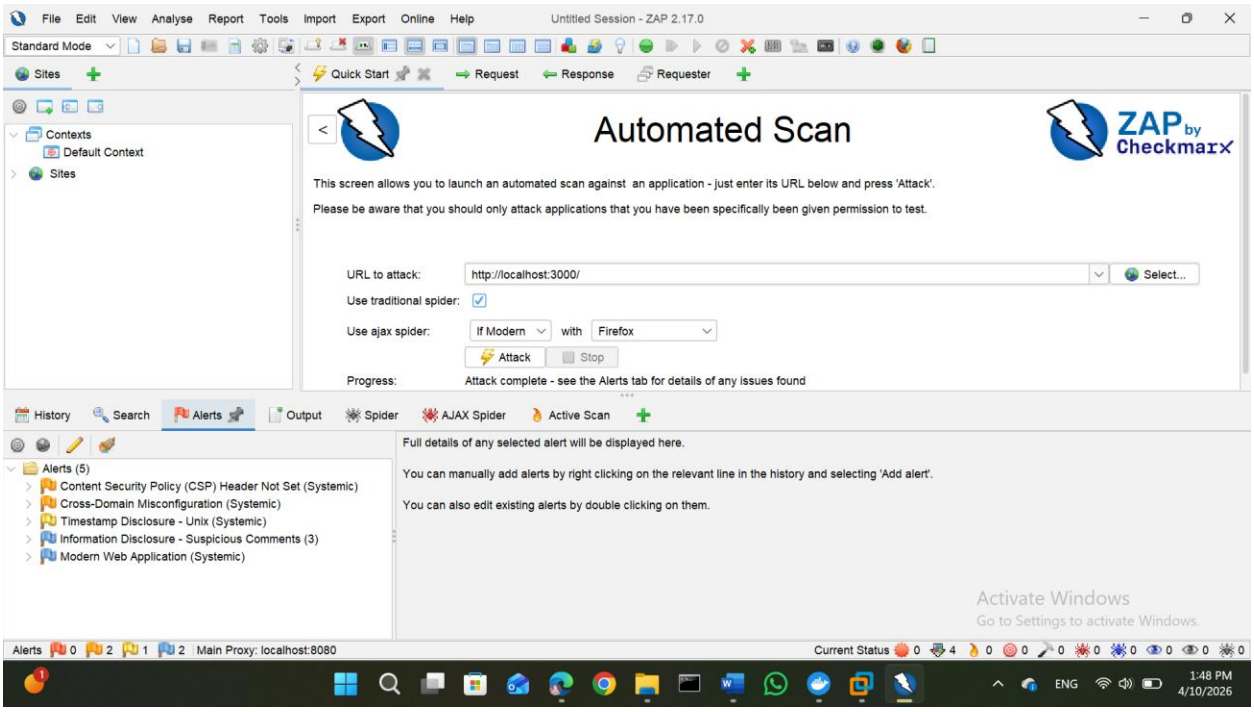
Client-side script execution confirmed



6. Automated Scan using OWASP ZAP

- The application was scanned using OWASP ZAP.
- Findings included:
  - Cross-Site Scripting vulnerabilities
  - Missing security headers
  - Information Disclosure

Multiple vulnerabilities detected



Areas of Improvement

1. Implement proper input validation and sanitization
2. Use parameterized queries to prevent SQL Injection
3. Enforce strong password policies
4. Add security headers such as Content Security Policy
5. Validate email formats properly

Conclusion

The application contains multiple security vulnerabilities including Cross-Site Scripting, SQL Injection, weak password policy, and improper input validation. These issues can be exploited by attackers to compromise the system. Proper security measures must be implemented to secure the application.